

# Vijay Singh

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## EDUCATION

**Indian Institute of Science, Bengaluru**  
*M.Tech in Transportation Systems Engineering*

*Aug 2024 – Jul 2026*  
CGPA: 7.8 / 10

**Punjab Technical University**  
*B.Tech in Civil Engineering*

*Aug 2016 – Jul 2020*  
CGPA: 8.8 / 10

## SKILLS

Python, R | ML | Gurobi, NetworkX, Google OR-Tools | VISSIM, SUMO | QGIS, Power BI | Pandas, NumPy, Matplotlib, Excel, LaTeX

## M.TECH THESIS

**Thesis Title:** Estimation of Origin-Destination Matrices Using Traffic Counts [\[Report\]](#) [\[Presentation\]](#)

**Importance:** Estimated OD matrices enable dynamic tracking of travel demand, early bottleneck identification, real-time signal timing and route optimization — building a continuously updating pipeline from junction-level camera data and replacing costly manual traffic surveys.

**About:** Turn counts from 6 junctions in the IISc corridor were fed into a VISSIM digital twin of 8 junctions, 16 zones, and 7 vehicle classes, with all simulation runs automated via a COM-based Python pipeline. Three OD estimation methods : MCP, SPSA, and TuRBO , were benchmarked head-to-head. MCP performed best, achieving  $R^2$  above 0.7 for Cars at high-volume junctions (~60–65% of network traffic) and keeping attraction sum deviations under 20%, while SPSA and TuRBO showed deviations up to 60% for some OD pairs.

## PROJECTS

### Branch-and-Cut Algorithm for GTSP

*Python, Gurobi, Matplotlib*

- Formulated the Symmetric Generalized Travelling Salesman Problem (GTSP) using node-selection and edge-selection variables.
- Implemented and compared baseline MILP, construction heuristic with 2-opt and Branch-and-Cut using subtour elimination constraints.

### Branch-and-Bound (B&B) for GAP and UFLP

*Python, Gurobi*

- Formulated the Generalized Assignment Problem (GAP) and Uncapacitated Facility Location Problem (UFLP)
- Developed custom B&B codes to solve LP files

### Resource-Constrained Shortest Path Problem (RCSP)

*Python, Gurobi, NetworkX*

- Built IP and LP link-flow formulations for RCSP with a distance budget constraint using Gurobi.
- Implemented Column Generation (CG) with path-flow variables, NetworkX, and label-correcting pricing.

### Branch-and-Cut (B&C) for Symmetric TSP

*Python, Gurobi, NetworkX, OR-Tools*

- Implemented a B&C framework for the Symmetric Travelling Salesman Problem (STSP) using connectivity cuts and 2-connected cuts.
- Compared results with and without cuts and benchmarked performance against Google OR-Tools.

### Contributor Verification using Machine Learning

*Python, ML*

- Developed a binary classification model to verify article contributors.
- Created count-based features and trained an XGBoost model with 10-fold CV, achieving around 91% validation accuracy.

### Last-Mile Mobility Data Analysis and Visualization

*Python, Excel, Power BI, QGIS*

- Analyzed metro last-mile survey data for Bengaluru, Delhi, and Nagpur by cleaning records and categorizing modes into NMT, IPT, PV, bus, and mixed modes.
- Developed Power BI dashboards and QGIS maps to visualize mode share, travel time, travel cost, distance, and station-level accessibility patterns.

## COURSES

Optimization Methods | Logistics & Freight Modeling | Traffic Network Equilibrium | Choice Modeling | Transportation Systems Modelling | Machine Learning for Cyber-Physical & Mobility Systems | Introduction to Multi-Modal Mobility Systems

## INTERNSHIPS

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### **TIA Project, IISc: Bagchi-Parthasarathy Hospital**

*Jun 2025 – July 2025*

- ANPR camera location planning, hospital site selection, data collection survey, draft bill preparation and VISSIM/SUMO simulation.

### **SJVN: Civil Engineering Intern**

*Apr 2023 – Jun 2023*

- BOQ and DPR review, quantity reconciliation, site measurement verification, and work progress reporting.

### **Deepak Builders: Road and Bridge Construction Intern**

*Apr 2020 – Jun 2020*

- Pathankot-Chakki Bank road and bridge construction stretch: site survey, material testing, mix checking, and field measurements.

### **CADD Centre, Pathankot:**

*Jan 2020 – Mar 2020*

- CAD software learning and project preparation using AutoCAD, 3ds Max, Revit and STAAD Pro.

## WORK EXPERIENCE

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### **Chegg: Subject Matter Expert - Freelance**

*Jul 2023 – Aug 2023*

- Prepared academic solutions for Civil Engineering and Psychology problems.

### **CADD Centre: Technical Software Trainer**

*Sep 2022 – Mar 2023*

- Trained students in CAD-based project work and conducted visiting classes at colleges.

### **Secretary to NCC Associate NCC Officer**

*Jul 2020 – Jul 2022*

- Assisted drill, parade, and training sessions, supported training camps with Indian Army and maintained records & documentation.

## ACHIEVEMENTS & ACTIVITIES

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- Attended the Large-Scale Optimization (LSO) Summer School at IIT Delhi in Jan 2026
- Attended the World Symposium on Sustainable Transport & Livability (WSSTL-2025) at IISc in Jun 2025
- Participated in Spectrum (Inter-Departmental Sports Meet at IISc) in 2025 & 2026, winning multiple medals
- Secured AIR 406 in the GATE 2024 Civil Engineering examination
- Secured University Rank 1 at Punjab Technical University for the academic session 2017
- Awarded the Best Manager Award by the Department of Management, CGC University, in 2020
- Awarded the Best Company Commander by NCC 23 Punjab Battalion in 2018

## Master of Technology

**NAME** : VIJAY SINGH **SR.NO** : 05-05-00-10-42-24-1-24024  
**DEPARTMENT** : Civil Engineering **SESSION** : 2024-25  
**TERM** : August-December 2024

| COURSE CODE | COURSE TITLE                     | CREDIT | GRADE |
|-------------|----------------------------------|--------|-------|
| CE 201      | Basic Geo-mechanics              | 3:0    | C     |
| CE 204      | Solid Mechanics                  | 3:0    | A     |
| CE 211      | Mathematics for Engineers        | 3:0    | B     |
| CE 217      | Fluid Mechanics                  | 3:0    | C     |
| CE 275      | Transportation Systems Modelling | 3:0    | B     |

|                             |              |          |          |
|-----------------------------|--------------|----------|----------|
| PERFORMANCE IN CURRENT TERM | CREDITS : 15 | GP : 105 | TGPA : 7 |
|-----------------------------|--------------|----------|----------|

Assistant Registrar  
Academic

\*\*Course marked with \* are substituted  
\*\*Course marked with # are repeated

\*This is a computer generated Term Transcript

## Master of Technology

**NAME** : VIJAY SINGH **SR.NO** : 05-05-00-10-42-24-1-24024  
**DEPARTMENT** : Civil Engineering **SESSION** : 2024-25  
**TERM** : January-April 2025

| COURSE CODE | COURSE TITLE                           | CREDIT | GRADE |
|-------------|----------------------------------------|--------|-------|
| CE 235      | Optimization Methods                   | 3:0    | B     |
| CE 262      | Public Transportation Systems Planning | 3:0    | B+    |
| CE 272      | Traffic Network Equilibrium            | 3:0    | D     |
| CE 271      | Choice Modeling                        | 3:0    | B     |
| CE 204A     | Pavement Engineering                   | 3:0    | A     |

|                               |              |          |            |
|-------------------------------|--------------|----------|------------|
| PERFORMANCE IN CURRENT TERM   | CREDITS : 15 | GP : 108 | TGPA : 7.2 |
| PERFORMANCE IN PREVIOUS TERMS | CREDITS : 15 | GP : 105 | CGPA : 7   |
| CUMULATIVE PERFORMANCE        | CREDITS : 30 | GP : 213 | CGPA : 7.1 |

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Academic

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\*\*Course marked with # are repeated

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## Master of Technology

**NAME** : VIJAY SINGH **SR.NO** : 05-05-00-10-42-24-1-24024  
**DEPARTMENT** : Civil Engineering **SESSION** : 2025-26  
**TERM** : August-December 2025

| COURSE CODE                   | COURSE TITLE                                             | CREDIT   | GRADE      |
|-------------------------------|----------------------------------------------------------|----------|------------|
| CP 219                        | Machine Learning for Cyber-Physical and Mobility Systems | 3:1      | C          |
| SL 221                        | Introduction to Multi-Modal Mobility Systems             | 3:1      | A          |
| SL 225                        | Logistics and Freight Modeling                           | 3:1      | B+         |
|                               |                                                          |          |            |
| PERFORMANCE IN CURRENT TERM   | CREDITS : 12                                             | GP : 92  | TGPA : 7.7 |
| PERFORMANCE IN PREVIOUS TERMS | CREDITS : 30                                             | GP : 213 | CGPA : 7.1 |
|                               |                                                          |          |            |
| CUMULATIVE PERFORMANCE        | CREDITS : 42                                             | GP : 305 | CGPA : 7.3 |

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Academic

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\*\*Course marked with # are repeated

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## Master of Technology

**NAME** : VIJAY SINGH **SR.NO** : 05-05-00-10-42-24-1-24024  
**DEPARTMENT** : Civil Engineering **SESSION** : 2025-26  
**TERM** : January-April 2026

| COURSE CODE                   | COURSE TITLE                                                                       | CREDIT       | GRADE    |
|-------------------------------|------------------------------------------------------------------------------------|--------------|----------|
| CE 299                        | Project : Origin Destination Matrices estimation using Simulation based Assignment | 0:22         | A        |
| PERFORMANCE IN CURRENT TERM   |                                                                                    | CREDITS : 22 | GP : 198 |
| PERFORMANCE IN PREVIOUS TERMS |                                                                                    | CREDITS : 42 | GP : 305 |
| CUMULATIVE PERFORMANCE        |                                                                                    | CREDITS : 64 | GP : 503 |

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Academic

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\*\*Course marked with # are repeated

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# GRADUATE APTITUDE TEST IN ENGINEERING 2024

अभियांत्रिकी स्नातक अभिक्षमता परीक्षा २०२४

ORGANISING INSTITUTE: INDIAN INSTITUTE OF SCIENCE, BENGALURU

## SCORE CARD

Name of the Candidate

**VIJAY SINGH**

Name of the Parent/Guardian

**PAWAN REKHA**

Registration No.

**CE24S38209033**

Test Paper

**Civil Engineering (CE)**

Date of Examination

**February 4, 2024**

GATE Score

**756**

\*Marks out of 100

**64.47**

All India Rank (AIR)  
in the test paper

**406**

**Qualifying Marks**

General

**28.3**

EWS/OBC-NCL

**25.4**

SC/ST/PwD

**18.8**

Number of candidates

**85869**

appeared for the test paper

\*Normalized marks across two sessions of the test paper



*Vijay Singh*

*Chandra Sekhar Seelamantula*

Prof. Chandra Sekhar Seelamantula  
Organising Chairperson, GATE 2024  
On behalf of NCB-GATE  
Ministry of Education (MoE)



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A candidate is considered **qualified** if the marks secured are greater than or equal to the qualifying marks mentioned for the category, for which a valid category certificate, if applicable, must be produced along with this Score Card.

This Score Card is valid  
up to 31<sup>st</sup> March 2027.

## GATE SCORE COMPUTATION

The GATE 2024 score is calculated using the formula

$$\text{GATE Score} = S_q + (S_t - S_q) \frac{(M - M_q)}{(M_t - M_q)}$$

where

M is the normalised marks obtained by the candidate in the paper mentioned on the GATE 2024 Score Card

$M_q$  is the qualifying marks for general category candidates in the paper

$M_t$  is the mean of marks of top 0.1% or top 10 (whichever is larger) of all the candidates who appeared for the test paper (i.e., including all sessions)

$S_q = 350$ , is the score assigned to  $M_q$

$S_t = 900$ , is the score assigned to  $M_t$

$M_q$  is 25 marks (out of 100) or  $\mu + \sigma$ , whichever is greater. Here  $\mu$  is the mean and  $\sigma$  is the standard deviation of marks of all the candidates who appeared for the test paper.



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अभियांत्रिकी स्नातक अभिक्षमता परीक्षा २०२४

ORGANISING INSTITUTE: INDIAN INSTITUTE OF SCIENCE, BENGALURU

## SCORE CARD

### COMPUTATION OF NORMALISED MARKS

Computer Science and Information Technology (CS) and Civil Engineering (CE) were conducted in two sessions in GATE 2024. For such multisession papers, a suitable normalisation is applied to take into account any variation in the difficulty levels of the question papers across sessions. The normalisation is done based on the assumption that, in multisession GATE papers, the distribution of the abilities of the candidates is nearly the same across sessions. This assumption is reasonable because the number of candidates appearing for the test papers is large, the number of candidates allotted to the sessions are comparable, and the procedure for allocation of candidates to the sessions is random.

The normalised marks of the  $j^{\text{th}}$  candidate in the  $i^{\text{th}}$  session, denoted by  $\hat{M}_{ij}$ , are computed as

$$\hat{M}_{ij} = \frac{\bar{M}_t^g - M_q^g}{\bar{M}_{ti} - M_{iq}} (M_{ij} - M_{iq}) + M_q^g$$

where

$M_{ij}$  is the actual marks obtained by the  $j^{\text{th}}$  candidate in the  $i^{\text{th}}$  session

$\bar{M}_t^g$  is the average marks of the top 0.1% of the candidates considering all sessions

$M_q^g$  is the sum of mean and standard deviation of marks of the candidates in the paper considering all sessions

$\bar{M}_{ti}$  is the average marks of the top 0.1% of the candidates in the  $i^{\text{th}}$  session and

$M_{iq}$  is the sum of the mean and standard deviation of marks in the  $i^{\text{th}}$  session.

Qualifying in GATE 2024 does not guarantee admission to a postgraduate program or scholarship/financial assistance. Admitting institutes may conduct additional tests or interviews for final selection of candidates.

Graduate Aptitude Test in Engineering (GATE) 2024 was organised by Indian Institute of Science, Bengaluru, on behalf of National Coordination Board (NCB) - GATE for the Department of Higher Education, Ministry of Education (MoE), Government of India.





भारत सरकार

Government of India



विजय सिंह

Vijay Singh

जन्म तिथि / DOB : 21/03/1998

पुरुष / Male



4049 1955 9092

आधार - आम आदमी का अधिकार



भारतीय विशिष्ट पहचान प्राधिकरण

Unique Identification Authority of India

पता:

आत्मज: दविन्द्र सिंह, गांव पक्का  
टियाला पोस्ट ओफिस कँडवाल तह  
नुरपुर, खंडवाल (३०), काँगड़ा,  
गढ़वाल, हिमाचल प्रदेश, 176201

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176201

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